

Chapter 5, Section 1

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July 27, 2026

1. Let C, D be any two divisors on a surface X , and let the corresponding invertible sheaves be $\mathcal{L}$ and $\mathcal{M}$. Show that

$$C.D = \chi(\mathcal{O}_X) - \chi(\mathcal{L}^{-1}) - \chi(\mathcal{M}^{-1}) + \chi(\mathcal{L}^{-1} \otimes \mathcal{M}^{-1}).$$

Proof. Applying the Riemann-Roch theorem to the divisor $-C - D$ gives

$$\chi(\mathcal{O}_X(-C - D)) = \frac{1}{2}(-C - D).(-C - D - K) + \chi(\mathcal{O}_X)$$

The first term on the right hand side simplifies to

$$(-C - D).(-C - D - K) = C^2 + 2C.D + D^2 + C.K + D.K.$$

Substituting back to above and isolating $C.D$ gives

$$\begin{aligned} C.D &= -\chi(\mathcal{O}_X) + \chi(\mathcal{O}_X(-C - D)) - \frac{1}{2}C.(C + K) - \frac{1}{2}D.(C + K) \\ &= \chi(\mathcal{O}_X) + \chi(\mathcal{O}_X(-C - D)) - \left(\chi(\mathcal{O}_X) - \frac{1}{2}C.(-C - K) \right) - \left(\chi(\mathcal{O}_X) - \frac{1}{2}D.(-D - K) \right) \\ &= \chi(\mathcal{O}_X) + \chi(\mathcal{O}_X(-C - D)) - \chi(\mathcal{O}_X(-C)) - \chi(\mathcal{O}_X(-D)). \end{aligned}$$

□

2. Let H be a very ample divisor on the surface X , corresponding to a projective embedding $X \subseteq \mathbb{P}^N$. If we write the Hilbert polynomial of X (III, Ex. 5.2) as

$$P(z) = \frac{1}{2}az^2 + bz + c,$$

show that $a = H^2$, $b = \frac{1}{2}H^2 + 1 - \pi$, where π is the genus of a nonsingular curve representing H , and $c = 1 + p_a$. Thus the *degree* of X in $\mathbb{P}^N$, as defined in (I, §7), is just H^2 . Show also that if C is any curve in X , then the degree of C in $\mathbb{P}^N$ is just $C.H$.

Proof. By the Riemann-Roch theorem, we have the equality

$$P(z) = \frac{1}{2}H^2z^2 - \frac{1}{2}(H.K)z + 1 + p_a.$$

Matching coefficients readily gives a, b , and c . We can rewrite the coefficient of z as

$$-\frac{1}{2}(H.K) = \frac{1}{2}H^2 - \frac{1}{2}H.(H + K),$$

and the adjunction formula (1.5) gives

$$2\pi - 2 = H.(H + K).$$

We conclude that

$$b = \frac{1}{2}H^2 + 1 - \pi.$$

If C is any curve in X , then the degree of C in $\mathbb{P}^N$ is just the degree of $\mathcal{O}_X(H) \otimes \mathcal{O}_C$ as a line bundle on C , which is equal to $C.H$ by (1.3). □

3. Recall that the *arithmetic genus* of a projective scheme D of dimension 1 is defined as $p_a(D) = 1 - \chi(\mathcal{O}_D)$ (III, Ex. 5.3).

- (a) If D is an effective divisor on a surface X , use (1.6) to show that $2p_a(D) - 2 = D \cdot (D + K)$.
- (b) $p_a(D)$ depends only on the linear equivalence class of D on X .
- (c) More generally, for *any* divisor D on X , we define the *virtual arithmetic genus* (which is equal to the ordinary arithmetic genus if D is effective) by the same formula: $2p_a - 2 = D \cdot (D + K)$. Show that for any two divisors C, D , we have

$$p_a(-D) = D^2 - p_a(D) + 2,$$

and

$$p_a(C + D) = p_a(C) + p_a(D) + C \cdot D - 1.$$

Proof.

- (a) We have the exact sequence

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_D \longrightarrow 0,$$

and taking Euler characteristics gives

$$\chi(\mathcal{O}_D) = \chi(\mathcal{O}_X) - \chi(\mathcal{O}_X(-D)).$$

On the other hand, the Riemann-Roch theorem gives

$$\chi(\mathcal{O}_X(-D)) = \frac{1}{2}D \cdot (D + K) + \chi(\mathcal{O}_X).$$

Substituting gives

$$1 - \chi(\mathcal{O}_D) = 1 + \frac{1}{2}D \cdot (D + K).$$

- (b) Follows from (a) and (1.1(4)), which states $D \cdot (D + K)$ depends only on the linear equivalence class of D on X .
- (c)

$$\begin{aligned} 2p_a(-D) - 2 &= -D \cdot (-D + K) \\ &= D \cdot (D - K) \\ &= 2D^2 - D \cdot (D + K) \\ &= 2D^2 - 2p_a(D) + 2 \\ 2p_a(C + D) - 2 &= (C + D) \cdot (C + D + K) \\ &= C \cdot (C + K) + D \cdot (D + K) + 2(C \cdot D) \\ &= 2p_a(C) - 2 + 2p_a(D) - 2 + 2(C \cdot D) \end{aligned}$$

□

- 4.** (a) If a surface X of degree d in $\mathbb{P}^3$ contains a straight line $C = \mathbb{P}^1$, show that $C^2 = 2 - d$.
(b) Assume $\text{char } k = 0$, and show for every $d \geq 1$, there exists a nonsingular surface X of degree d in $\mathbb{P}^3$ containing the line $x = y = 0$.

Proof.

- (a) By (Ex. 1.5), $C^2 = -C \cdot K - 2$. By (1.3), $C \cdot K = d - 4$, because C is a straight line and $\mathcal{O}_X(K) = \mathcal{O}_X(d - 4)$. Hence, $C^2 = 2 - d$.
- (b) Fix an integer $d > 0$, let L be any line in $\mathbb{P}^3$ with ideal sheaf $\mathcal{I}$. Consider the linear system $\mathfrak{d}$ on $\mathbb{P}^3$ associated to $H^0(\mathbb{P}^3, \mathcal{I}(d))$. It has base-locus equal to L , and by Bertini's theorem (III, 10.9) (III, 10.9.1) (III, 10.9.2) (III, Ex. 11.3), a general member of $\mathfrak{d}$ is connected and can have singularities only at the base points. Now,

□

5. (a) If X is a surface of degree d in $\mathbb{P}^3$, then $K^2 = d(d-4)^2$.
 (b) If X is a product of two nonsingular curves C, C' , of genus g, g' respectively, then $K^2 = 8(g-1)(g'-1)$.

Proof.

- (a) The adjunction formula gives $\mathcal{O}_X(K) = \mathcal{O}_X(d-4)$. If H is a general hyperplane section of X in $\mathbb{P}^3$, then $K = (d-4)H$, and $H^2 = d$ by (Ex. 1.2). Hence,

$$K^2 = (d-4)^2 H^2 = d(d-4)^2.$$

- (b) By (II, Ex. 8.3), there is an isomorphism $\omega_X \cong p_1^* \omega_C \otimes p_2^* \omega_{C'}$. In particular, if K_C (resp. $K_{C'}$) is a canonical divisor of C (resp. C'), then $p_1^* K_C + p_2^* K_{C'}$ is a canonical divisor of X . Denote $D = p_1^* K_C$ and $D' = p_2^* K_{C'}$. The support of D consists of $2g-2$ copies of C' which are the fibres of the points in K_C with respect to $p_1 : X \rightarrow C$. It follows that

$$D \cdot D' = (\deg K_C) \cdot (\deg K_{C'}) = 4(g-1)(g'-1).$$

Also, for any finite set of points in C , there exists a canonical divisor with disjoint support. The intersection pairing only depends on the linear equivalence class, so we have $D^2 = 0$. We conclude that $K^2 = 2D \cdot D' = 8(g-1)(g'-1)$.

□

6. (a) If C is a curve of genus g , show that the diagonal $\Delta \subseteq C \times C$ has self-intersection $\Delta^2 = 2 - 2g$. (Use the definition of ω_C in (II, §8).)
 (b) Let $l = C \times \text{pt}$ and $m = \text{pt} \times C$. If $g \geq 1$, show that l , m , and Δ are linearly independent in $\text{Num}(C \times C)$. Thus $\text{Num}(C \times C)$ has rank ≥ 3 , and in particular, $\text{Pic}(C \times C) \neq p_1^* \text{Pic } C \oplus p_2^* \text{Pic } C$.

Proof.

- (a) By definition, $\omega_C \cong \mathcal{O}_X(-\Delta) \otimes \mathcal{O}_\Delta$, and by (1.3), $\deg_\Delta(\mathcal{O}_X(-\Delta) \otimes \mathcal{O}_\Delta) = -\Delta^2$.
 (b)

□

7. Algebraic Equivalence of Divisors. Let X be a surface. Recall that we have defined an algebraic family of effective divisors on X , parametrized by a nonsingular curve T , to be an effective Cartier divisor D on $X \times T$, flat over T (III, 9.8.5). In this case, for any two closed points $0, 1 \in T$, we say that the corresponding divisors D_0, D_1 on X are prealgebraically equivalent. Two arbitrary divisors are prealgebraically equivalent if they are differences of prealgebraically equivalent divisors. Two divisors D, D' are *algebraically equivalent* if there is a finite sequence $D = D_0, D_1, \dots, D_n = D'$ with D_i and D_{i+1} prealgebraically equivalent for each i .

- (a) Show that the divisors algebraically equivalent to 0 form a subgroup of $\text{Div } X$.
- (b) Show that linearly equivalent divisors are algebraically equivalent.
- (c) Show that algebraically equivalent divisors are numerically equivalent.

Proof.

- (a)
- (b) We divide the problem into steps by successively proving more general cases, i.e., when D is an (1) irreducible curve, (2) effective divisor, and (3) arbitrary divisor.
 - (1) By definition, there exists a rational function $f \in K(X)^*$ such that $D - D' = (f)$. On an open subset $U \subset X$, let g, g' be locally defining equations of D, D' such that $g = fg'$. The divisors associated to $h = (1-t)g + tg'$ for each $t \in \mathbb{A}^1$ gives a family of divisors on $U \times \mathbb{A}^1$. Covering C by open subsets and patching local equations give an effective Cartier divisor E on $X \times T$, flat over T , and $E_0 = D, E_1 = D'$. Hence, if D and D' are linearly equivalent irreducible curves in X , then they are algebraically equivalent.
- (c)

□

8. Cohomology Class of a Divisor. For any divisor D on the surface X , we define its cohomology class $c(D) \in H^1(X, \Omega_X)$ by using the isomorphism $\text{Pic } X \cong H^1(X, \mathcal{O}_X^*)$ of (III, Ex. 4.5) and the sheaf homomorphism $d \log : \mathcal{O}_X^* \rightarrow \Omega_X$ (III, Ex. 7.4c). Thus we obtain a group homomorphism $c : \text{Pic } X \rightarrow H^1(X, \Omega_X)$. On the other hand, $H^1(X, \Omega_X)$ is dual to itself by Serre duality (III, 7.13), so we have a nondegenerate bilinear map

$$\langle \ , \ \rangle : H^1(X, \Omega_X) \times H^1(X, \Omega_X) \rightarrow k.$$

- (a) Prove that this is compatible with the intersection pairing, in the following sense: for any two divisors D, E on X , we have

$$\langle c(D), c(E) \rangle = (D.E) \cdot 1$$

in k .

- (b) If $\text{char } k = 0$, use the fact that $H^1(X, \Omega_X)$ is a finite-dimensional vector space to show that $\text{Num } X$ is a finitely generated free abelian group.

Proof.

- (a) We need to first show numerically equivalent divisors give isomorphic cohomology classes. Suppose $D \equiv 0$ and $D \neq 0$ in $\text{Pic } X$. Let (U, f) be a locally defining equation of D . We want to show $df/f = 0$. Let P be any point in X ,
- (b)

□

9. (a) If H is an ample divisor on the surface X , and if D is any divisor, show that

$$(D^2)(H^2) \leq (D.H)^2.$$

- (b) Now let X be a product of two curves $X = C \times C'$. Let $l = C \times \text{pt}$, and $m = \text{pt} \times C'$. For any divisor D on X , let $a = D.l$, $b = D.m$. Then we say D has *type* (a, b) . If D has type (a, b) , with $a, b \in \mathbb{Z}$, show that

$$D^2 \leq 2ab,$$

and equality holds if and only if $D = bl + am$.

Proof.

- (a) Before we prove the statement, let us look at an example first. Consider the quadric surface $Q = \mathbb{P}^1 \times \mathbb{P}^1$ with H divisor of type $(1, 1)$, which gives the standard embedding into $\mathbb{P}^3$. If D be a divisor of type (a, b) , then by (1.4.3), the left-hand side is equal to $2(a^2 + b^2)$, while the right-hand side is $(a + b)^2$ (IV, 6.4.1) (Ex. 1), so the inequality holds.

Now, let D be any divisor, and let E be any irreducible curve in X . By replacing H with mH for some $m \gg 0$ if necessary, we may assume that H is very ample. The hypothesis of (1.2) is satisfied, so further assume H is irreducible, nonsingular, and meets E and each irreducible component of D transversally.

We will show that the inequality holds for D if and only if it holds for $D + E$ (IV, 1.3). Plugging in $D + E$ into above simplifies to

$$(D^2)(H^2) + 2(D.C)(H^2) \leq (D.E)^2 + 2(D.E)(C.E),$$

so it is enough to show

$$(D.E)(H^2) \leq (D.H)(E.H)$$

for any divisor D and irreducible curve E . Here, we can apply the reasoning above again, and for the last time further reduce to the case D is also an irreducible curve.

(b)

□

- 10.** *Weil's Proof [2] of the Analogue of the Riemann Hypothesis for Curves.* Let C be a curve of genus g defined over the finite field $\mathbb{F}_q$, and let N be the number of points of C rational over $\mathbb{F}_q$. Then $N = 1 - a + q$, with $|a| \leq 2g\sqrt{q}$. To prove this, we consider C as a curve over the algebraic closure k of $\mathbb{F}_q$. Let $f: C \rightarrow C$ be the k -linear Frobenius morphism obtained by taking q th powers, which makes sense since C is defined over $\mathbb{F}_q$, so $X_q \cong X$ (IV, 2.4.1). Let $\Gamma \subseteq C \times C$ be the graph of f , and let $\Delta \subseteq C \times C$ be the diagonal. Show that $\Gamma^2 = q(2 - 2g)$, and $\Gamma.\Delta = N$. Then apply (Ex. 1.9) to $D = r\Gamma + s\Delta$ for all r and s to obtain the result.

- 11.** In this problem, we assume that X is a surface for which $\text{Num } X$ is finitely generated.

- (a) If H is an ample divisor on X , and $d \in \mathbb{Z}$, show that the set of effective divisors D with $D.H = d$, modulo numerical equivalence, is a finite set.
- (b) Now let C be a curve of genus $g \geq 2$, and use (a) to show that the group of automorphisms of C is finite, as follows. Given an automorphism σ of C , let $\Gamma \subseteq X = C \times C$ be its graph. First show that if $\Gamma \equiv \Delta$, then $\Gamma = \Delta$, using the fact that $\Delta^2 < 0$, since $g \geq 2$ (Ex. 1.6). Then use (a).

- 12.** If D is an ample divisor on the surface X , and $D' \equiv D$, then D' is also ample. Give an example to show, however, that if D is very ample, D' need not be very ample.

Proof.

□